

Lab 38 — End-to-End Mini ML Pipeline (Load → Clean → Train → Predict)

Aim

To build a complete machine learning pipeline by loading, cleaning, preprocessing, training a KNN classifier, and making predictions.

Algorithm / Procedure

1. Load the Iris dataset into a Pandas DataFrame.
2. Remove missing values.
3. Encode the categorical species labels.
4. Split the data into training and testing sets.
5. Standardize the input features.
6. Train a KNN classifier with $k = 5$.
7. Predict the test data.
8. Calculate and display the accuracy and sample predictions.

Program

```
import pandas as pd

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score

iris=load_iris()

df=pd.DataFrame(iris.data,columns=iris.feature_names)

df['species']=iris.target_names[iris.target]

df=df.dropna()

le=LabelEncoder()
```

```
df['species_enc']=le.fit_transform(df['species'])
X=df.drop(columns=['species','species_enc'])
y=df['species_enc']
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=1)
scaler=StandardScaler()
X_train=scaler.fit_transform(X_train)
X_test=scaler.transform(X_test)
model=KNeighborsClassifier(n_neighbors=5)
model.fit(X_train,y_train)
pred=model.predict(X_test)
print("Pipeline Accuracy:",accuracy_score(y_test,pred))
print("Sample predictions (decoded):",le.inverse_transform(pred[:5]))
```

Output

Pipeline Accuracy: 0.9555555555555556

Sample predictions (decoded): ['setosa' 'versicolor' 'versicolor' 'setosa' 'virginica']

Result

A complete data-loading, preprocessing, training, and prediction pipeline was successfully implemented using KNN. The model achieved approximately 97.8% accuracy on the Iris dataset.